

2021-Ph-H1-Q5

Section: Our Dynamic Universe

Topic: Forces

A graph shows terminal velocities for two objects, P and Q.

Which statements about motion are correct?

Solution:

- Object P reaches terminal velocity = 50 m/s (✓)
- Object Q also reaches constant velocity (✓)
- Object Q travels **less**, not more distance (X)

Answer: B

Guidance for Students:

Terminal velocity = when upward and downward forces are balanced. Area under the v–t graph shows distance.

Revision Tip:

If terminal velocity is lower, total distance travelled is often less — check by comparing areas under the curves!